

9- Mini Hotel Reservation System

Description

A Mini Hotel Reservation System manages rooms' reservations and customer reviews.

- Hotel rooms are a 2D array (floors × rooms).
- Customers book or cancel rooms.
- Customers view room availability.
- Save bookings to a file.
- Admin can manage customer's reservations and reviews.
- Previous bookings are loaded automatically on startup.

Structs should be created for:

Customer:

CustomerID, Name, Email, UserName, Password, CreditCardNumber, RoomNumber, StartDate, EndDate

Room:

RoomNumber, CustomerNumber, BookingStatus

Review:

ReviewID, RoomNumber, ReviewContent

Admin:

AdminID, UserName, Password

Customer Functions

- **Log in:** Authenticate using username and password.
- **Sign up:** Create a new customer account.
- **Check room availability:** submit the floor number and startDate along with endDate, and its available rooms are shown.
- **Reserve room:** Submit room number, startDate, endDate, and creditCardNumber (in case of previously paid reservation).
- **Add Review:** submit review with the room number.
- **Log out:** Exit the customer session.

	Admin Functions ● Log in: Authenticate using admin credentials. ● Add Reservation: Submit room number, startDate, endDate, and customer ID. ● View Room Reviews: View all reviews, unseen reviews, and seen reviews for the last 30 days. ● Edit Reservation: Change the start and end date of reservation for customer ID (if applicable). ● Log out: Exit the admin session.
	Program Running Scenario The program should ask the user whether he is: 1. Customer 2. Admin
	If the user chooses Customer: The customer logs in using his username and password. After login, a list of choices is displayed to him to choose the function he would like to perform.
	If the user chooses Admin: After login, a list of choices is displayed to him to choose the function he would like to perform.
	Struct and Arrays Declaration You should declare arrays for Customers, Rooms, Reviews, and Admins, with predefined records included according to the minimum delivery requirements.

Deliverables	The submitted program must include: ● Well-structured and organized code. ● Login functionality for Admin and Customer. ● All the functions that are assigned to the Admin and Customer. ● Automatic calculation of the last 30 days' reviews. ● Automatic calculation for available rooms. Minimum Requirements at Delivery: ● At least 5 predefined Customers. ● At least 50 predefined Rooms (10 at each floor). ● At least 5 predefined Reviews. ● At least 1 Admin account.
Bonus	● GUI (Graphical User Interface) ● Any extra functionality (check with the supervisor first)
Very Important Note	You will use the files only in 2 places: 1) To save the data before you close the project. 2) To load the data into arrays, variables & structs after starting the project. It is not allowed to use files for data processing. You must use arrays, variables and structs to process (input, display, edit) any data while running the project.